

Md Monirul Islam

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Education

- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
• *B.Sc. in Electrical and Electronic Engineering* March 2018 – May 2023
CGPA: 3.85/4.00 (**3.98/4.00 in the final year**), ranked **25 of 189**, awarded with Honours
Relevant Courses: Electrical & Electronic Circuits, Calculus I & II, Ordinary & Partial Differential Equations, Linear Algebra, Communication System, Digital Signal Processing, Digital Electronics, Power Electronics, Control Systems, Microprocessors and Embedded Systems, Analog Integrated Circuits and Design, VLSI Circuits and Design

Research Interests

- Robotics • Autonomous Systems • Multi-Robot Coordination • Motion Planning • Path Planning • Real-Time Control • Sensor Fusion • Embedded Systems • Robot Perception • Human-Robot Interaction • Field Deployment of Robotic Platforms

Research Contributions

- **Decentralized spatiotemporal coordination for multi-robot fleets.** A fully distributed A* path-planning protocol in which each robot publishes its plan as a sequence of timestamped grid waypoints, and every peer treats those plans as spatiotemporal reservations subject to three conflict checks (node conflict, edge swap, final-cell occupation). A continuous timestamp-maintenance step keeps published ETAs honest between re-planning events; obstacles are handled without dropping the active plan. Validated by months of zero-collision production operation on a 3-robot fleet in a live human-occupied factory. *Manuscript in preparation, first / corresponding author.*
- **Bilateral cross-coupled BLDC speed synchronization on a dual-MCU architecture.** A distributed speed-synchronization scheme where each of two BLDC motors is regulated by its own microcontroller running a PID loop with velocity and acceleration feedforward; a bilateral cross-coupled term, evaluated identically on both nodes from a dedicated inter-MCU speed exchange, drives the inter-motor speed difference toward zero. All gains GA-tuned offline against an identified motor model. Outperforms an independent-PID baseline on both transient and steady-state synchronization metrics. *Manuscript under review, first / corresponding author.*
- **Closed-form analytical IK with FK-anchored goal accumulation for a 7-DOF mobile manipulator.** A from-scratch analytical inverse-kinematics solver replacing MoveIt's KDL — deterministic, branch-stable, constant-time. Smooth turret-scaling near the home-pose singularity preserves continuity through the boundary; an FK-anchored goal accumulator re-anchors the commanded position to the FK-actual wrist pose each tick, substantially reducing un-commanded cross-axis motion. Verified by a unit-test suite covering workspace sweeps, joint-limit clipping, branch consistency at boundaries, and FK round-tripping.
- **Latency-compensated 5-state EKF for visual-fiducial AGV localization.** Per-robot pose fused from wheel-encoder odometry and AprilTag camera fixes. The state includes velocity-bias terms that absorb systematic encoder error over long corridors; camera updates are applied at their true measurement timestamp via a rewind-and-replay step on a bounded encoder history buffer. Two operating modes (Kalman filter while moving, complementary filter while parked) seamlessly hand off through arrival / departure events.
- **Fault-tolerant dual-network feedback communication architecture** providing redundant pathways for coordinating unmanned multi-node robotic platforms in the field. *Manuscript under review, co-author.*
- **Browser-side Pareto optimizer for heavy-lift multirotor sizing.** Published interactive tool that enumerates feasible series-parallel battery configurations against a motor's measured thrust / current / throttle curve, evaluating hover-throttle reachability, per-motor current limits, voltage compatibility, and cruise margin. Runs entirely client-side; designed as a reproducible companion artifact to the heavy-lift drone project.
- **α -Graphyne ballistic nanoribbon FET.** Undergraduate research with Dr. Mahbub Alam modelling quantum transport in 2D-material field-effect transistors. *Published in Micromachines (MDPI), 2023.*

Publications

- **Md Monirul Islam** et al., "Peer-Aware Distributed Path Planning for a Production-Deployed Multi-AGV Fleet in a Human-Occupied Garment Factory", In Preparation, 2026 (first / corresponding author).
- **Md Monirul Islam** et al., "Experimental Speed Synchronization of a Dual-BLDC Tracked Differential-Drive Vehicle Using a GA-Tuned Bilateral Cross-Coupled Controller on a Dual-MCU Architecture", Under Review, 2026 (first / corresponding author).
- **Md Monirul Islam** et al., "A Dual-Network Feedback Communication System for Multi-Node Unmanned Robotic Control", Under Review, 2026 (co-author).
- Md Shakil Tanvir, Fiaz Ahmed Tonmoy, **Md Monirul Islam**, Mainul Islam, Mahamudul Hasan, Raihan Ur Rashid Razeen, "Optimizing Master-Slave Broadcast Communication in Multi-Node Networks Using RS485 Standard", 2024, 27th International Conference on Computer and Information Technology (ICCIT), IEEE. doi: [10.1109/ICCIT64611.2024.11022430](https://doi.org/10.1109/ICCIT64611.2024.11022430)

- H. Khan, **Md Monirul Islam**, R.I. Roya, S.N. Azad, “Performance Analysis of an α -Graphyne Nano-Field Effect Transistor”, *Micromachines*, MDPI, 2023. doi: [10.3390/mi14071385](https://doi.org/10.3390/mi14071385)

Professional Experience

- **Research and Development Unit, Cybernetics Hi-Tech Solutions (Pvt) Ltd** Dhaka, Bangladesh
R&D Engineer and Team Lead (June 2023 – present)
 - Lead the R&D engineering team taking robotic platforms from concept through field deployment; own architecture, coding, hardware integration, commissioning, and post-deployment support.
 - **CyberFleet (production AGV fleet)**. Lead engineer end-to-end. The platform productised the work and is now operating continuously at The Urmi Group RMG facility, Dhaka.
 - **Jontro Soinik v1 / v2 EOD ROVs**. Owned the electrical / firmware side across both generations. v1 was gifted to the Peruvian Armed Forces by the Bangladesh Army and is deployed under the UN Peacekeeping Mission in Mali (2024); v2 was Army-validated (disrupter and IP55 tests) and is set for Congo (2025). Ran on-site operator-training sessions and field hand-over to receiving units.
 - **Jontro Soinik 2.0 (7-DOF mobile manipulator)**. Owned the control / software stack for the tracked tele-op platform.
 - **Heavy-lift multirotor (50 kg payload)**. Owned BMS, propulsion sizing, frame, and power-distribution PCBs; shipped the sizing study as an interactive in-browser tool on the portfolio site.
 - **SD15-N target-tracking surveillance drone**. Designed the autonomous-tracking pipeline (vision → MAVLink → Pixhawk).
 - **Titan 550 industrial CoreXY 3D printer**. Two-iteration product build; the in-house fab loop that produces fixtures, brackets, and prototypes for every other platform on this list.
 - **In-house PCB & CNC**. Designed all controller / communication boards across the ROV & AGV platforms in Altium / KiCad and milled them in-house on a 3-axis CNC — no external fab.
 - **Other**: hybrid VTOL quad-plane prototype; real-time pet-food sorting line for an industrial-automation client.
- **Ulkasemi Pvt. Limited** Dhaka, Bangladesh
Intern (Dec 2022 – Mar 2023)
 - Worked on IC design using TSMC 180 nm and GPDK 90 nm technologies.
 - Developed a Wishbone interface and validated a high-speed BiFET Op-Amp in Verilog.
- **Walton Hi-Tech Industries Limited** Dhaka, Bangladesh
Industrial Trainee (Nov 2022)
 - Observed and analyzed PCB manufacturing and EV assembly workflows.
 - Gained exposure to industrial electronics production workflows.

Selected Undergraduate Projects

- **Real-time traffic speed estimation with YOLOv2**: Trained YOLOv2 in MATLAB to detect vehicles and estimate per-vehicle speed from monocular highway video.
- **6-DOF robotic arm (analytical IK)**: Designed in SOLIDWORKS, 3D-printed entirely on a self-built hobby printer using custom 3D-printed planetary gearboxes; closed-form analytical inverse-kinematics control on stepper motors.
- **Autonomous GPS-waypoint delivery drone**: Mechanical design in SOLIDWORKS, Arduino flight controller with PID stabilisation, GPS-waypoint navigation, and a custom payload-deployment mechanism.
- **Self-built hobby 3D printer**: Designed every component in SOLIDWORKS, fabricated the frame on a laser cutter, Arduino Mega controller running custom Marlin firmware. The workhorse that produced every other undergraduate project’s parts.
- **RV32I multicycle RISC-V processor + Wishbone memory controller**: Custom Verilog implementation with full testbench verification; physical-layout exploration in Cadence at TSMC 180 nm.
- **OMR machine from discrete digital logic**: Optical-mark recognition built entirely from discrete NAND / RAM / MUX / Comparator / LDR components — no microcontroller in the loop.
- **Regulated boost converter (TSMC 180 nm, Cadence)**: Schematic capture and physical layout of a voltage-regulated DC-DC boost converter.
- **Closed-loop induction-motor speed control**: V/f scalar control with SVPWM inverter modulation in MATLAB / Simulink.
- **Particle-concentration light-scattering sensor**: Laser + photodetector light-scattering setup with MATLAB-side calibration curves and signal-conditioning circuit.
- **Real-time environmental monitoring**: Solar-powered IoT node logging temperature, humidity, and CO₂ over GSM; Arduino + Autodesk Eagle PCB.
- **Laboratory variable DC power supply**: Linear-regulator design with current limiting and short-circuit protection; MATLAB efficiency study.
- **Electrical-services design (seven-storey building)**: End-to-end electrical design: load estimation, panel scheduling, cable sizing, lighting / power layouts, riser diagram, earthing scheme.

Skills Summary

- **Robotics & Middleware:** ROS 2 (Jazzy), MoveIt 2, ros2.control, JointTrajectoryController, URDF / Xacro / SRDF, AprilTag, MAVLink, dronekit, pymavlink
- **Embedded & MCU:** STM32 (HAL & LL), ESP32, AVR, Arduino, ARM Cortex-M, µVision Keil, BigTreeTech 32-bit boards
- **Control & Algorithms:** Closed-form analytical IK, Extended Kalman Filter, PID (cascaded, GA-tuned), S-curve motion profiling, distributed A* path planning, peer-aware spatiotemporal reservations
- **Flight Control & Vision:** ArduPilot, iNav, Betaflight, Pixhawk, YOLOv11n, OpenCV
- **Simulation & Design Software:** MATLAB / Simulink, Cadence Virtuoso, PSPICE, PSAF, Icarus Verilog, GTKWave
- **PCB & Circuit Design:** Altium Designer, KiCad, Autodesk Eagle, Proteus; in-house 3-axis CNC milling for prototype boards
- **CAD & Mechanical Design:** SolidWorks
- **3D Printing & CNC:** Marlin (customised), Ultimaker Cura, Creality CR-6 SE; FlatCAM, Universal G-code Sender, GRBL
- **Programming Languages:** Python, C, Embedded C, Verilog
- **Documentation:** Microsoft Office Suite, L^AT_EX, Jekyll (al-folio)
- **Soft Skills:** Leadership, Project Management, Field Commissioning, Operator Training, Technical Writing

Honors and Awards

- University Merit Scholarship, BUET (three times — January 2019, January 2020, January 2022)
 - Awarded to students with outstanding academic record (GPA 3.90+) in a semester
- Dean's List Award, BUET (four times — all academic years)
 - Awarded to students obtaining an average GPA of 3.75+ in two consecutive semesters of an academic year
- University Stipend (three times — January 2018, July 2018, July 2021)
- Higher Secondary Certificate (HSC-2017) Board Scholarship
- Runners-up, Bangabandhu Creative Talent Hunt 2014, Pabna Division

Extra-curricular Activities

- Affiliate Member, BUET Robotics Society (BRS). (Jan 2022 – Mar 2023)
- Affiliate Member, Team Interplanetar – BUET Mars Rover Team. (Jun 2022 – Mar 2023)

Field Deployments

- 3-robot AGV fleet – live production at The Urmi Group RMG facility, Dhaka, 2025 – present
- Jontro Soinik v1 EOD ROV – UN Peacekeeping Mission, Mali, 2024 (Bangladesh Army → Peruvian Armed Forces handover)
- Jontro Soinik v2 EOD ROV – tested by Bangladesh Army, set for UN Peacekeeping deployment in the Republic of Congo, 2025

References

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